

PROJECT SYNOPSIS

SmartVital: A Unified IoT-Integrated Multi-Disease Early Detection System

for Heart Attack, Diabetes, and Lung Cancer using Machine Learning and Deep Learning

Field	Details
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1. Introduction

The global burden of non-communicable diseases (NCDs) such as cardiovascular disorders, diabetes mellitus, and lung cancer continues to rise at an alarming pace. Early and accurate detection of these conditions is the single most impactful strategy for improving patient survival rates and reducing healthcare costs. However, most existing detection systems are either single-disease focused, require expensive clinical setups, or produce "black-box" predictions that clinicians cannot interpret or trust.

This project proposes SmartVital — a unified, intelligent health monitoring system that simultaneously detects risk levels for three critical diseases: heart attack / heart disease, diabetes, and lung cancer. The system integrates real-time IoT sensor data with trained Machine Learning (ML) and Deep Learning (DL) models, and incorporates Explainable AI (XAI) to ensure every prediction is transparent and justifiable.

2. Problem Statement

The following critical gaps exist in current disease detection research and systems:

- Most ML/DL systems are single-disease focused — no unified multi-disease monitoring system exists at an affordable, deployable level.
- Existing models trained on clean, balanced datasets (e.g., Breast Cancer Wisconsin) achieve 95%+ accuracy with minimal effort, leaving no room for genuine research contribution.
- Deep learning models are "black boxes" — clinicians cannot understand why a patient was flagged as high risk, reducing clinical adoption.

- No student-level research has combined IoT hardware + multi-disease ML models + explainability (XAI) in a single unified system.

This project directly addresses all four gaps.

3. Objectives

1. To collect and preprocess multi-disease datasets from Kaggle covering heart attack, diabetes, and lung cancer.
2. To train baseline ML models and progressively improve accuracy using advanced DL architectures built from scratch.
3. To apply Explainable AI (SHAP / LIME) to make every model prediction interpretable and clinically trustworthy.
4. To integrate IoT sensors (Raspberry Pi + Arduino) for real-time vital sign collection feeding directly into the trained models.
5. To build a unified real-time risk dashboard providing Low / Medium / High risk scores for all three diseases simultaneously.
6. To document the full accuracy improvement pipeline as a research contribution suitable for IEEE/Springer publication.

4. Dataset Details

Four carefully selected Kaggle datasets are used, each chosen for complementary strengths and research challenges:

4.1 Heart Disease Detection — Dataset 1 (1000 Patients)

Attribute	Detail
Source	Kaggle
Sample Size	1,000 patient records
Key Features	Age, Gender, Cholesterol, Blood Pressure, Heart Rate, Smoking, Alcohol Intake, Exercise, Family History, Diabetes, Obesity, Stress Level, Blood Sugar, Exercise-Induced Angina, Chest Pain Type
Target Column	HeartDisease (0 = No, 1 = Yes)
Chest Pain Types	ATA (Atypical Angina), NAP (Non-Anginal Pain), ASY (Asymptomatic), TA (Typical Angina)
Research Challenge	ASY (Asymptomatic) patients feel no symptoms yet have heart disease — hardest to detect

4.2 Heart Disease Detection — Dataset 2 (900 Patients)

Attribute	Detail
Source	Kaggle (Cleveland-based)

Attribute	Detail
Sample Size	~900 patient records
Key Features	Age, Sex, ChestPainType, RestingBP, Cholesterol, FastingBS, RestingECG, MaxHR, ExerciseAngina, Oldpeak, ST_Slope
Target Column	HeartDisease (0 = No, 1 = Yes)
Notable Features	Oldpeak (ST depression — higher value = higher risk), ST_Slope (Up/Flat/Down — Down = highest risk)
Usage	Cross-validation and ensemble model training alongside Dataset 1

4.3 Early Heart Attack Prediction (5000+ Patients)

Attribute	Detail
Source	Kaggle
Sample Size	5,000+ patient records
Key Features	id, gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, bmi, smoking_status
Target Column	stroke (0 = No stroke, 1 = Stroke/Heart Attack)
Research Value	Largest dataset — used for final DL model training and generalization testing
Challenge	Class imbalance — strokes are rare events in the population

4.4 Diabetes Detection (700+ and 1900+ Patients)

Attribute	Detail
Source	Kaggle (PIMA Indians Diabetes — extended versions)
Sample Sizes	700+ records (Version 1) and 1,900+ records (Version 2)
Key Features	Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age
Target Column	Outcome (0 = Non-diabetic, 1 = Diabetic)
IoT Mapping	Glucose level, Blood Pressure, BMI — directly measurable via wearable sensors
Strategy	Train on 700-record version first; validate on 1900-record version for robustness

4.5 Lung Cancer Risk Prediction (1000 Patients)

Attribute	Detail
Source	Kaggle
Sample Size	1,000 patient records

Attribute	Detail
Key Features	Age, Gender, Air Pollution, Alcohol Use, Dust Allergy, Occupational Hazards, Genetic Risk, Chronic Lung Disease, Balanced Diet, Obesity, Smoking, Passive Smoker, Chest Pain, Coughing of Blood, Fatigue, Weight Loss, Shortness of Breath, Wheezing, Swallowing Difficulty, Clubbing of Finger Nails, Frequent Cold, Dry Cough, Snoring
Scoring System	All symptom/risk features rated on 1–8 scale (1 = minimal, 8 = severe)
Target Column	Level: Low / Medium / High cancer risk (3-class classification)
Research Challenge	3-class classification with symptom severity scales — baseline accuracy ~62–68%

5. Proposed Methodology

Phase 1 — Data Collection & Preprocessing

- Download all 5 datasets from Kaggle
- Handle missing values using median/mode imputation
- Encode categorical variables (e.g., ChestPainType, Gender, Smoking Status)
- Apply SMOTE (Synthetic Minority Oversampling Technique) for imbalanced datasets
- Feature selection using Chi-square test and correlation matrix to retain most predictive columns

Phase 2 — Baseline ML Model Training

- Train Logistic Regression as starting baseline — document initial accuracy (~62–78% depending on dataset)
- Train Random Forest and XGBoost — expected improvement to ~80–85%
- Train Support Vector Machine (SVM) for comparison
- Each model evaluated using: Accuracy, Precision, Recall, F1-Score, ROC-AUC

Phase 3 — Deep Learning Model (Custom, From Scratch)

- Build custom Artificial Neural Network (ANN) using TensorFlow / Keras
- Architecture: Input Layer → Dense(128, ReLU) → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.2) → Output Layer
- For lung cancer (3-class): Output uses Softmax activation; for binary tasks: Sigmoid activation
- Train with Adam optimizer, monitor validation loss with EarlyStopping callback
- Target accuracy: 88–92% across all disease models

Phase 4 — Explainable AI (XAI) Integration

- Apply SHAP (SHapley Additive exPlanations) to identify which features most influence each prediction

- Apply LIME (Local Interpretable Model-agnostic Explanations) for individual patient-level explanations
- Generate feature importance visualizations: "Why was this patient flagged as HIGH risk?"
- This makes the system clinically trustworthy — doctors can see the reasoning, not just the result

Phase 5 — IoT Hardware Integration

Sensor	Measures	Maps To Disease	Cost (Approx.)
MAX30102	Heart Rate + SpO2 (blood oxygen)	Heart Attack Detection	Rs. 250-400
AD8232 ECG Module	Heart electrical signal (ECG)	Heart Attack / Arrhythmia	Rs. 400-600
DHT11 / DHT22	Body temperature + humidity	Lung Disease / Fever	Rs. 100-150
GSR Sensor	Skin conductance / sweat response	Diabetes stress correlation	Rs. 150-250
Blood Pressure Module	Systolic / Diastolic BP	Heart + Diabetes features	Rs. 400-600
Raspberry Pi 4 / Arduino Uno	Central processing + data relay	All three diseases	Rs. 2000-4000

Phase 6 — Unified Dashboard & Alert System

- Build real-time web dashboard using Streamlit (Python)
- Live sensor readings displayed alongside model-predicted risk scores
- Simultaneous output for all three diseases: Heart Risk | Diabetes Risk | Lung Risk
- Color-coded alerts: Green (Low) | Yellow (Medium) | Red (High)
- Email/SMS alert triggered automatically when any disease risk crosses HIGH threshold

6. Accuracy Improvement Strategy

The following table shows the planned accuracy progression — this progression IS the research contribution:

Disease	Baseline (Logistic Reg.)	After XGBoost	After Custom DL	After XAI + Tuning
Heart Disease (1000)	~76-78%	~84-86%	~89-91%	~91-93%
Heart Disease (900)	~74-76%	~82-85%	~87-90%	~90-92%
Early Heart Attack (5000+)	~72-75%	~80-83%	~86-89%	~88-91%

Disease	Baseline (Logistic Reg.)	After XGBoost	After Custom DL	After XAI + Tuning
Diabetes (700+)	~72-74%	~80-82%	~85-88%	~87-90%
Diabetes (1900+)	~74-76%	~82-84%	~87-90%	~89-92%
Lung Cancer (1000)	~62-65%	~75-78%	~83-86%	~86-89%

7. Innovation & Research Contribution

This project is novel at the student research level for the following reasons:

- **Multi-Disease Unified System:** No existing affordable system simultaneously detects heart attack, diabetes, and lung cancer risk in real time — this is the primary novelty.
- **From-Scratch Training on Hard Datasets:** All models are trained from scratch (no pretrained models) on challenging datasets with low baselines, making the accuracy improvement a genuine contribution.
- **Explainable AI Layer:** SHAP + LIME integration makes the system clinically interpretable — addressing the #1 reason DL models are not adopted in hospitals.
- **IoT Hardware Integration:** Physical sensor data feeds into trained models in real time — combining embedded systems with AI in a single deployable prototype.
- **Multi-Dataset Validation:** Using 5 datasets across 3 diseases with varying sizes (700 to 5000+ samples) ensures the models are robust, not overfitted.

8. Tools & Technologies

Component	Tool / Technology
Programming Language	Python 3.x
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn
Machine Learning	Scikit-learn (Logistic Regression, Random Forest, XGBoost, SVM)
Deep Learning	TensorFlow 2.x / Keras
Explainable AI	SHAP, LIME
Class Imbalance Handling	imbalanced-learn (SMOTE)
Dashboard / Frontend	Streamlit
Development Environment	Google Colab (training) + VS Code (integration)
IoT Hardware	Raspberry Pi 4 / Arduino Uno + sensors
Dataset Source	Kaggle / UCI ML Repository
Version Control	GitHub

9. Project Timeline

Week	Phase	Milestone
Week 1-2	Research & Setup	Literature review, dataset download, environment setup, EDA
Week 3-4	Preprocessing	Data cleaning, missing value handling, SMOTE, feature selection for all 5 datasets
Week 5-6	Baseline ML Models	Train Logistic Regression, Random Forest, XGBoost — document initial accuracies
Week 7-8	Deep Learning	Build and train custom ANN from scratch — push accuracy to 85%+
Week 9	Explainable AI	Integrate SHAP + LIME — generate feature importance charts for all models
Week 10	IoT Integration	Connect sensors to Raspberry Pi, test live data collection
Week 11	Dashboard	Build Streamlit dashboard with live predictions and alert system
Week 12	Report & Presentation	Final documentation, result compilation, mentor presentation

10. Expected Outcomes

- A fully trained multi-disease ML/DL system achieving 88–92% accuracy across all three disease categories.
- An explainable AI layer (SHAP + LIME) that justifies every high-risk prediction with feature-level reasoning.
- A working IoT prototype (Raspberry Pi + sensors) that collects real-time vitals and feeds them into the trained models.
- A unified Streamlit dashboard providing simultaneous risk scores for heart attack, diabetes, and lung cancer.
- A complete research paper draft suitable for submission to IEEE Access, IEEE Sensors Journal, or Springer Journal of Medical Systems.

11. References (Datasets)

1. Heart Disease Dataset (1000 patients) — Kaggle: kaggle.com/datasets/rashadrmammadov/heart-disease-prediction
2. Heart Disease Dataset (Cleveland-based, ~900 patients) — Kaggle: kaggle.com/datasets/fedesoriano/heart-failure-prediction
3. Early Heart Attack Prediction (5000+ patients, stroke dataset) — Kaggle: kaggle.com/datasets/fedesoriano/stroke-prediction-dataset
4. PIMA Indians Diabetes Dataset (700+ and 1900+ patients) — Kaggle: kaggle.com/datasets/uciml/pima-indians-diabetes-database
5. Lung Cancer Risk Prediction (1000 patients) — Kaggle: kaggle.com/datasets/thedevastator/cancer-patients-and-air-pollution-a-new-link

This synopsis is submitted for approval as part of the Summer Internship Research Project.

[Student Signature] _____ [Mentor Approval] _____